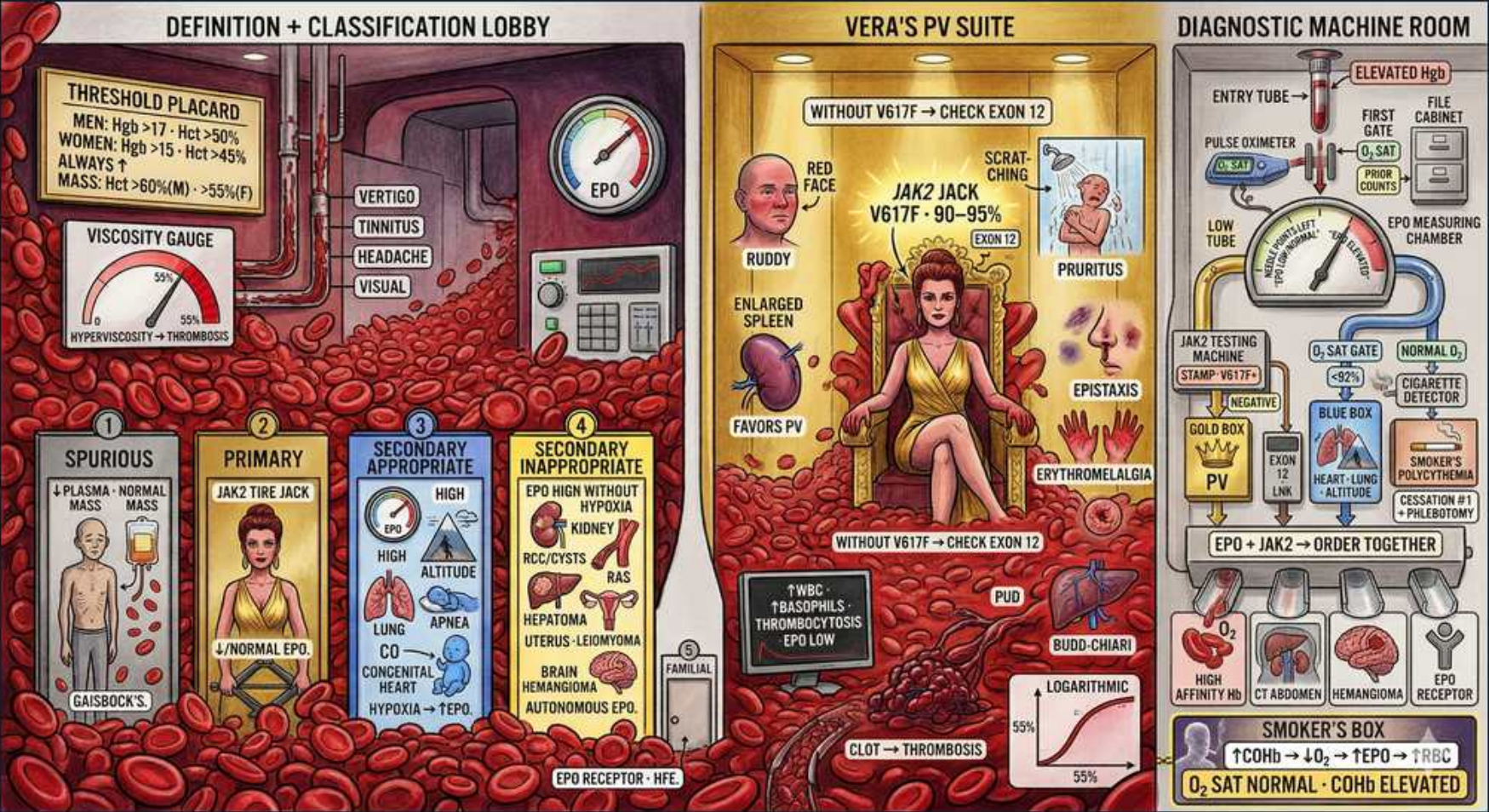


Polycythemia & Polycythemia Vera Workup



1. Hb/Hct thresholds

WHAT YOU SEE

Men Hb>16.5 Hct>49, Women Hb>16 Hct>48

WHAT IT ENCODES

Polycythemia (erythrocytosis) is defined by sex-specific thresholds: men Hgb >16.5 g/dL or Hct >49%, women Hgb >16 g/dL or Hct >48%. These are the 2016 WHO criteria for screening; true polycythemia vera (PV) major criterion now requires Hgb >16.5 (men) / >16 (women) or Hct >49% / >48%, or elevated red cell mass. The first branch point is always distinguishing a true increase in red cell mass from a relative (spurious) rise from plasma contraction.

BOARD TRAP

WRONG: 'A single high Hct on a dehydrated ER patient confirms polycythemia vera.' Apparent/relative polycythemia from plasma volume contraction (vomiting, diuretics, burns) raises Hct without raising true red cell mass. Discriminator: rehydrate and recheck, and check serum EPO + JAK2 before labeling any erythrocytosis as PV.

2. Hyperviscosity sx

WHAT YOU SEE

Viscosity gauge: vertigo, tinnitus, headache, visual

WHAT IT ENCODES

Increased red cell mass raises whole-blood viscosity, producing headache, vertigo, tinnitus, blurred/visual disturbances, and lethargy. The dominant danger is thrombosis (arterial and venous) — the leading cause of death in PV. Symptoms worsen sharply once Hct climbs above ~55-60%, which is why the treatment target is Hct <45%.

BOARD TRAP

WRONG: 'Hyperviscosity symptoms in this polycythemic patient mean Waldenström macroglobulinemia.' Hyperviscosity from IgM paraprotein (Waldenström) presents similarly but with a monoclonal IgM spike and normal/low Hct, not erythrocytosis. Discriminator: in PV the viscosity is driven by RBC mass (high Hct); in Waldenström it is driven by IgM (check SPEP/serum viscosity).

3. Spurious (Gaisböck)

WHAT YOU SEE

1 SPURIOUS: ↓plasma, normal mass, Gaisböck's

WHAT IT ENCODES

Relative/spurious (apparent) polycythemia: plasma volume is low while true red cell mass is normal, so Hct rises artifactually. Gaisböck syndrome is the classic stress/smoking, hypertensive, overweight middle-aged man with mildly elevated Hct. Causes include dehydration, diuretic overuse, and chronic stress. EPO and JAK2 are normal; red cell mass study (if done) is normal.

BOARD TRAP

WRONG: 'Mildly elevated Hct in a hypertensive smoking businessman warrants hydroxyurea for PV.' Gaisböck is a relative polycythemia with normal red cell mass — cytoreduction is not indicated. Discriminator: normal serum EPO and absent JAK2 with normal red cell mass; manage by stopping smoking/diuretics and controlling volume status.

4. Primary = low EPO

WHAT YOU SEE

2 PRIMARY: JAK2, low EPO

WHAT IT ENCODES

Primary polycythemia = polycythemia vera, a JAK2-driven myeloproliferative neoplasm where erythroid precursors proliferate autonomously, independent of EPO. The marrow makes RBCs without being told to, so the body suppresses EPO → serum EPO is LOW. Low EPO + JAK2 mutation is the diagnostic signature. Often accompanied by leukocytosis and thrombocytosis (panmyelosis).

BOARD TRAP

WRONG: 'PV should show a high serum EPO because the patient is making so many red cells.' EPO is LOW in PV precisely because production is autonomous and feedback-suppressed. High EPO instead points to a secondary cause (hypoxia or EPO-secreting tumor). Discriminator: low EPO favors PV, high EPO essentially excludes it.

5. Secondary appropriate

WHAT YOU SEE

3 SECONDARY APPROPRIATE: high EPO, hypoxia (altitude, COPD, OSA, cyanotic heart)

WHAT IT ENCODES

Appropriate secondary polycythemia: chronic tissue hypoxia is correctly sensed and drives a high EPO and a compensatory rise in RBC mass. Causes: high altitude, COPD, obstructive sleep apnea, right-to-left shunt/cyanotic congenital heart disease, chronic CO exposure (smoking), and high-affinity hemoglobin variants. EPO is HIGH and JAK2 is negative.

BOARD TRAP

WRONG: 'High Hct plus splenomegaly in a COPD patient confirms PV.' COPD/OSA cause appropriate secondary erythrocytosis with HIGH EPO and no JAK2 mutation; splenomegaly is uncommon. Discriminator: hypoxemia (low SaO2/PaO2) + high EPO + negative JAK2 = secondary, not PV.

6. Secondary inappropriate

WHAT YOU SEE

4 SECONDARY INAPPROPRIATE: EPO high without hypoxia (renal, hepatoma, leiomyoma, cerebellar hemangioblastoma)

WHAT IT ENCODES

Inappropriate secondary polycythemia: EPO is high WITHOUT hypoxia because a tumor autonomously secretes EPO. Classic sources: renal cell carcinoma, hepatocellular carcinoma, cerebellar hemangioblastoma, uterine leiomyoma, and pheochromocytoma. Mnemonic for paraneoplastic EPO: 'Really High Hematocrit Lifts Pressure' (Renal cell, Hepatocellular, Hemangioblastoma, Leiomyoma, Pheo). EPO is HIGH, SaO2 normal, JAK2 negative.

BOARD TRAP

WRONG: 'Erythrocytosis with high EPO and normal oxygen saturation must be PV.' Normal SaO2 with HIGH EPO points to an EPO-secreting tumor (inappropriate secondary), not PV (which has LOW EPO). Discriminator: image the kidneys/liver/posterior fossa for a tumor; PV has low EPO + JAK2+.

7. JAK2 V617F

WHAT YOU SEE

JAK2 V617F 90-95%

WHAT IT ENCODES

JAK2 V617F is a gain-of-function point mutation (valine→phenylalanine at codon 617) in the JH2 pseudokinase autoinhibitory domain, causing constitutive JAK-STAT signaling and EPO-independent erythroid growth. Present in ~95% of PV (and ~50-60% of essential thrombocythemia and primary myelofibrosis). It is the major molecular criterion for PV diagnosis.

BOARD TRAP

WRONG: 'A positive JAK2 V617F by itself confirms polycythemia vera.' JAK2 V617F also occurs in essential thrombocythemia and primary myelofibrosis, so it is not specific for PV. Discriminator: interpret JAK2 together with Hgb/Hct, low EPO, and bone marrow morphology (panmyelosis) to land on PV specifically.

8. V617F neg → exon 12

WHAT YOU SEE

Without V617F → check exon 12

WHAT IT ENCODES

If JAK2 V617F is negative but PV is still suspected (low EPO, high Hct), test for JAK2 exon 12 mutations, found in ~3-4% of PV. Exon 12 PV tends to present with isolated erythrocytosis (less leukocytosis/thrombocytosis) and younger patients. Together V617F + exon 12 account for ~98-99% of PV cases.

BOARD TRAP

WRONG: 'A negative JAK2 V617F rules out polycythemia vera.' About 3-4% of PV is V617F-negative but carries a JAK2 exon 12 mutation. Discriminator: in V617F-negative patients with low EPO and erythrocytosis, reflex to exon 12 testing before excluding PV.

9. Pruritus

WHAT YOU SEE

Aquagenic pruritus after warm shower

WHAT IT ENCODES

Aquagenic pruritus — intense itching triggered by warm water/showering, without a rash — is a hallmark constitutional symptom of PV, attributed to mast cell/basophil degranulation and histamine release. It can be debilitating and is a strong clinical clue. May respond to antihistamines, SSRIs (paroxetine), phototherapy, or to cytoreduction/JAK inhibitor (ruxolitinib).

BOARD TRAP

WRONG: 'Itching after a warm shower with a high Hct is just dry skin/eczema.' Aquagenic pruritus is a recognized PV feature and should prompt EPO + JAK2 testing, not just emollients. Discriminator: itching with NO primary skin lesion + erythrocytosis points to PV (or another myeloproliferative neoplasm).

10. Erythromelalgia

WHAT YOU SEE

Erythromelalgia: burning red hands/feet

WHAT IT ENCODES

Erythromelalgia = burning pain with erythema and warmth of the hands/feet from platelet-mediated microvascular occlusion in PV (and essential thrombocythemia). It is a microvascular thrombotic symptom and classically responds dramatically to low-dose aspirin (81 mg). PV also carries macrovascular thrombosis risk (stroke, MI, DVT/PE).

BOARD TRAP

WRONG: 'Burning red painful extremities in a polycythemic patient is peripheral neuropathy — start gabapentin.' Erythromelalgia is platelet-driven microvascular ischemia, not neuropathy, and responds to low-dose aspirin. Discriminator: relief with aspirin and association with thrombocytosis/PV distinguishes it from neuropathic pain.

11. Splenomegaly / ruddy

WHAT YOU SEE

Enlarged spleen, ruddy face, favors PV

WHAT IT ENCODES

Splenomegaly (from extramedullary hematopoiesis/sequestration) and facial plethora/ruddy cyanosis strongly favor PV over secondary or relative polycythemia, which do not enlarge the spleen. Splenomegaly is part of the WHO clinical picture; progressive massive splenomegaly may signal transformation to post-PV myelofibrosis.

BOARD TRAP

WRONG: 'Splenomegaly with erythrocytosis is most likely from chronic hypoxia/COPD.' Secondary polycythemia (hypoxic or tumor-driven) does NOT cause splenomegaly; an enlarged spleen points to a myeloproliferative neoplasm (PV). Discriminator: spleen + plethora + low EPO + JAK2 = PV, not secondary.

12. Low EPO + JAK2

WHAT YOU SEE

EPO + JAK2 order together; low EPO + JAK2+ = PV

WHAT IT ENCODES

The high-yield workup move: order serum EPO and JAK2 together. The diagnostic combination LOW serum EPO + JAK2 mutation (V617F or exon 12) establishes PV with high confidence and usually obviates red cell mass studies. A normal/high EPO redirects you to secondary causes. WHO PV diagnosis = major criteria (Hgb/Hct or RBC mass, marrow panmyelosis, JAK2) + minor criterion (subnormal EPO).

BOARD TRAP

WRONG: 'You must measure red cell mass with isotope studies before diagnosing PV.' In practice, low EPO + JAK2 mutation reliably confirms PV; formal red cell mass is rarely needed. Discriminator: reserve red cell mass studies for ambiguous cases where EPO/JAK2 don't fit.

13. Budd-Chiari

WHAT YOU SEE

PV → Budd-Chiari thrombosis

WHAT IT ENCODES

PV is the leading occult cause of Budd-Chiari syndrome (hepatic vein thrombosis) and of other splanchnic/unusual-site thromboses (portal, mesenteric, cerebral venous). A young patient with Budd-Chiari and a 'normal' or only mildly elevated CBC (masked by hemodilution from ascites/bleeding) should be screened with JAK2 V617F.

BOARD TRAP

WRONG: 'This Budd-Chiari patient has a normal Hct, so a myeloproliferative neoplasm is excluded.' Splanchnic thrombosis can mask erythrocytosis via plasma expansion/GI bleeding, yet JAK2-positive PV is a top cause. Discriminator: test JAK2 V617F in any unprovoked splanchnic-vein thrombosis regardless of CBC.

14. CO / smoker's box

WHAT YOU SEE

Smoker's box: ↑COHb, ↓O₂, normal SaO₂, elevated COHb

WHAT IT ENCODES

Heavy smoking causes secondary polycythemia via carbon monoxide: CO binds hemoglobin (carboxyhemoglobin), reducing O₂ delivery and driving a compensatory EPO rise. Pulse oximetry reads falsely NORMAL because standard pulse ox cannot distinguish COHb from oxyHb — you need co-oximetry (ABG) to reveal the elevated COHb and true low O₂ content.

BOARD TRAP

WRONG: 'Normal pulse oximetry rules out a hypoxic/CO cause of this smoker's polycythemia.' Standard pulse ox misreads carboxyhemoglobin as oxyhemoglobin, so SpO₂ looks normal despite impaired O₂ delivery. Discriminator: measure carboxyhemoglobin by co-oximetry; an elevated COHb explains 'smoker's polycythemia' with normal SpO₂.

15. Treatment: Hct<45

WHAT YOU SEE

In the diagnostic machine room, a treatment placard/target gauge set to Hct <45% with phlebotomy tubing, an aspirin tablet, and a hydroxyurea bottle — the PV management bundle.

WHAT IT ENCODES

PV treatment lowers thrombotic risk: PHLEBOTOMY to a target Hct <45% (the CYTO-PV trial showed fewer cardiovascular deaths/thrombosis at <45% vs 45-50%), plus LOW-DOSE ASPIRIN (81 mg) for all patients unless contraindicated. Add CYTOREDUCTION (first-line hydroxyurea; ruxolitinib for hydroxyurea-resistant/intolerant) for high-risk patients — age >60 or prior thrombosis. Control cardiovascular risk factors; PV can transform to post-PV myelofibrosis or acute leukemia.

BOARD TRAP

WRONG answer: targeting a 'normal' Hct around 50% or withholding aspirin in PV. Discriminator: the evidence-based target is Hct <45% (CYTO-PV), and aspirin plus phlebotomy is standard even in low-risk patients — reserve hydroxyurea for high-risk (age >60 or prior clot), not for everyone.